Chapter 4 Breakout (Pt. I)

STAT 4714

Questions

1. In a gambling game, a woman is paid \$3 if she draws a jack or a queen and \$5 if she draws a king or an ace from an ordinary deck of 52 playing cards. If she draws any other card, she loses. How much money should she expect to make after pulling one card?

2. The density function of the proportion of EE students who watched the last VT football game is

$$f(x) = \frac{4}{\pi(1+x^2)}, \quad 0 < x < 1.$$

   Find the expected value of X.

3. The density function of X, the number of hours you will spend on the next homework, is

$$f(x) = \begin{cases} x, & 0 < x < 1. \\ 2 - x, & 1 \leq x < 2. \end{cases}$$

   Find $E(X)$.

4. Suppose that X and Y have the following joint probability function:

$f(x,y)$		x = 2	x = 4
y	1	0.10	0.15
	3	0.20	0.30
	5	0.10	0.15

   (a) Find the expected value of $g(X,Y) = XY^2$.

   (b) Find μ_X and μ_Y.

1. In a gambling game, a woman is paid \$3 if she draws a jack or a queen and \$5 if she draws a king or an ace from an ordinary deck of 52 playing cards. If she draws any other card, she loses. How much money should she expect to make after pulling one card?

$X \equiv$ money received (in \$)

$x = 0, 3, 5$

$P(X=3) = P(\text{pulling jack or queen}) = 8/52$

$P(X=5) = P(\text{pulling king or ace}) = 8/52$

$P(X=0) = P(\text{pulling something else}) = 36/52$

$E(X) = 0 \cdot P(X=0) + 3 \cdot P(X=3) + 5 \cdot P(X=5)$

$= 3(8/52) + 5(8/52) = \$1.23$

2. The density function of the proportion of EE students who watched the last VT football game is

$$f(x) = \frac{4}{\pi(1+x^2)}, \quad 0 < x < 1.$$

Find the expected value of X.

$$E(X) = \int_0^1 x f(x)\,dx = \int_0^1 x \frac{4}{\pi(1+x^2)}\,dx$$

$u = 1+x^2$

$du = 2x\,dx$

$\Rightarrow x\,dx = \frac{1}{2}du$

$$= \frac{4}{\pi}\int_0^1 \frac{x}{1+x^2}\,dx$$

$$= \frac{4}{\pi}\int_1^2 u^{-1}\,\frac{1}{2}du$$

$$= \frac{2}{\pi}\left[\log(u)\Big|_1^2\right] = \underline{\frac{2\log(2)}{\pi}}$$

3. The density function of X, the number of hours you will spend on the next homework, is

$$f(x) = \begin{cases} x, & 0 < x < 1. \\ 2 - x, & 1 \leq x < 2. \end{cases}$$

Find $E(X)$.

$$E(X) = \int_x x f(x)\,dx = \int_0^1 x^2\,dx + \int_1^2 (2x - x^2)\,dx$$

$$= \frac{1}{3}x^3 \Big|_0^1 + \left[x^2 - \frac{1}{3}x^3\right]_1^2$$

$$= \frac{1}{3} + \left(4 - \frac{8}{3} - \left(1 - \frac{1}{3}\right)\right)$$

$$= \underline{1}$$

4. Suppose that X and Y have the following joint probability function:

$f(x,y)$		x = 2	x = 4
y	1	0.10	0.15
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(a) Find the expected value of $g(X,Y) = XY^2$.

(b) Find μ_X and μ_Y.

(a) $E(XY^2) = \sum_x \sum_y xy^2 f(x,y)$

$= 2(1)^2(0.1) + 2(3)^2(0.20) + 2(5)^2(0.10)$

$+ 4(1)^2(0.15) + 4(3)^2(0.30) + 4(5)^2(0.15)$

$= \underline{35.2}$

(b) $E(X) = \sum_x x f(x) = 2(0.1+0.2+0.1) + 4(0.15+0.3+0.15)$

$= \underline{3.2}$

$E(Y) = \sum_y y f(y) = 1(0.1+0.15) + 3(0.2+0.3) + 5(0.1+0.15)$

$= \underline{3}$